

DEEPAK SINGH

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New Delhi, India

SUMMARY

Aspiring AI/ML Engineer and competitive programmer with a strong passion for problem-solving, algorithms, and building large-scale recommendation systems. Deeply interested in Machine Learning and the emerging world of Agentic & Generative AI. Outside of work, I maintain a steady fitness routine.

EDUCATION

Guru Gobind Singh Indraprastha University (USAR)

New Delhi, India

Bachelor of Technology in Artificial Intelligence and Data Science (CGPA: 8.62/10.0)

2024 – 2028

TECHNICAL SKILLS

Languages : Python, C++, SQL

Machine Learning & AI: PyTorch, Hugging Face, Transformers, LangChain, LangGraph, RAG, NLP, Instruction Fine-Tuning

Data Science : NumPy, Pandas, Matplotlib, Hugging Face Datasets

Visualization & Frontend : React, Streamlit

Backend & DevOps : FastAPI, MongoDB, Git, MySQL

Core Concepts : Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networking

EXPERIENCE

ML Intern, Indian Army (505 Army Base Workshop)

June 2026 – Aug 2026

Under Lt. Col. Narender Pal Singh Mann | Delhi Cantonment, Delhi

- Developed a **predictive engine health monitoring system** for T-72 battle tanks using ensemble learning and anomaly detection, achieving **92% accuracy** in early fault detection — shifting maintenance from reactive to proactive and reducing downtime by **30%**.
- Engineered an end-to-end time-series pipeline processing **12+ sensor parameters** (temperature, vibration, pressure) using Pandas, NumPy, and other visualization libraries.
- Collaborated directly with defense engineers to translate operational pain points into ML-driven solutions, influencing maintenance strategies across hundreds of tanks with the DGM.

PROJECTS

Movie Recommendation System | Python, Scikit-learn, NLTK, Streamlit [\[GitHub\]](#) [\[Demo\]](#) Mar 2026 – May 2026

- Built a **content-based recommender** on the **TMDB 5000** dataset using genres, keywords, cast, crew, and overview.
- Applied **NLP preprocessing** (tokenization, stopwords removal, PorterStemmer) and **TF-IDF vectorization** to compute a **cosine similarity matrix** across 4,800+ movies.
- Deployed a **Streamlit web app** on Render returning **top-5 similar movies** with live posters via the **TMDB API**.
- Tech Stack**: Python, Pandas, NumPy, Scikit-learn, NLTK, Streamlit, TMDB API, Git LFS

WhatsApp Chat Analyzer | Python, Pandas, Regex, Matplotlib, Streamlit [\[GitHub\]](#) [\[Demo\]](#) Jul 2025 – Sep 2025

- Built a **Streamlit web app** that parses exported WhatsApp chats (.txt) using **regex** into a structured Pandas DataFrame.
- Enabled **user-level and date-range filtering** to generate personalized analytics for any participant or group.
- Visualized insights including **most busy users, word clouds, most common words, and monthly/hourly activity charts** using Matplotlib.
- Tech Stack**: Python, Pandas, NumPy, Regex, Matplotlib, Seaborn, WordCloud, Streamlit, Render

AWARDS & CERTIFICATIONS

- 2nd Position, Smart Delhi Ideation 2024** — Awarded by **Hon'ble Lt. Governor of Delhi, Shri Vinai Kumar Saxena** at Vigyan Bhawan. Ranked 2nd among 1000+ teams in a GGSIPU-organized innovation challenge.
- 2nd Runner-Up, Smart India Hackathon 2025** — Built an AI-powered Train Traffic Control system maximizing throughput using ML and data science.